

Annotated TCAD Command Code

Portfolio-ready annotated version of SDE, SDevice, GIDL, LDD/Halo, and SONOS Program/Erase command scripts

Source: user-provided TCAD_COMMANDCODE.docx. This annotated document is intended for portfolio explanation and code review. Keep the original command files as the executable source.

Document Map

1. LONG SDE CODE
2. LONG SDEVICE transfer_curve CODE
3. LONG SDEVICE output_CURVE CODE
4. LONG GIDL SDE CODE
5. LONG GIDL SDEVICE TRANSFERCURVE CODE
6. LONG GIDL OUTPUTCURVE CODE
7. SHORT SDE CODE
8. SHORT TRANSFER CODE
9. DEVICE SETUP AND PHYSICAL MODELS
10. LDD+HALO SDE CODE
11. PROGRAM/ERASE SDEVICE CODE
12. SPROCESS IV sdevice

1. LONG SDE CODE

- Builds the long-channel reference MOSFET structure.
- Defines substrate/gate regions, source/drain doping, contacts, mesh refinement, and mirror symmetry.

Annotated Code

```
; Clear any previously defined SDE geometry, doping, mesh, and contacts.
(sde:clear)
; Define the process-up direction so vertical coordinates are interpreted consistently.
(sde:set-process-up-direction "+z")

; Select doping species and carrier models based on whether the device is nMOS or pMOS.
(if (string=? "@Type@" "nMOS")
  (begin
    ; Define the substrate dopant species.
    (define DopSub "BoronActiveConcentration")
    ; Define the source/drain and gate dopant species.
    (define DopSD "ArsenicActiveConcentration")
    ; Set the background substrate doping concentration.
    (define SubDop 1e18 )
  )
  (begin
    ; Define the substrate dopant species.
    (define DopSub "ArsenicActiveConcentration")
    ; Define the source/drain and gate dopant species.
    (define DopSD "BoronActiveConcentration")
    ; Set the background substrate doping concentration.
    (define SubDop 1e18)
  )
)

; Gate Length parameter imported from Sentaurus Workbench.
(define Lg @lgate@)
; Total simulated lateral length, including gate and source/drain extension region.
(define Ltot (+ Lg (* 2 0.01) 0.8) )

; Substrate thickness used for the 2D device cross-section.
(define Hsub 1.0)
; Gate oxide thickness parameter imported from Workbench.
(define Tox @Toxide@)
; Polysilicon gate height.
(define Hpol 0.2)

; Minimum mesh size control used near critical regions.
(define Gpn 0.005)

; Approximate junction depth used for source/drain implants.
(define Xj 0.1 )

; Derived dimensions are computed from the primary process/device parameters.
;Derived quantities
; Half-device lateral boundary for building a symmetric structure.
(define Xmax (/ Ltot 2.0))
; Half gate length.
(define Xg (/ Lg 2.0))

; Bottom substrate coordinate.
(define Ysub Hsub)
; Gate oxide top/bottom coordinate based on oxide thickness.
(define Ygox (* Tox -1.0))
```

```

; Top polysilicon coordinate.
(define Ypol (- Ygox Hpol))

(define Lcont Xmax)

; Set the default Boolean operation for overlapping SDE geometry creation.
(sdegeo:set-default-boolean "ABA")

; Create a 2D rectangular material region such as silicon, oxide, spacer, or polysilicon.
(sdegeo:create-rectangle
  (position 0.0 Ysub 0.0 )
  (position Xmax 0.0 0.0 )
; Name the silicon body/substrate region.
  "Silicon" "R.Substrate"
)

(sdegeo:create-rectangle
  (position 0.0 0.0 0.0 )
  (position Xg Ygox 0.0 )
; Name the gate oxide region.
  "Oxide" "R.Gateox"
)

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xg Ypol 0.0 )
; Define the re-oxidized oxide region around the polysilicon gate.
  "Oxide" "R.PolyReox"
)

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xg Ypol 0.0 )
; Define the polysilicon gate region.
  "PolySi" "R.Polygate"
)

; Assign electrical contacts to selected geometry edges.
(sdegeo:set-contact
  (find-edge-id (position (* (+ Xmax Xg) 0.5) 0.0 0.0))
; Label this contact as the drain.
  "drain")
(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ypol 0.0))
; Label this contact as the gate.
  "gate")
(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ysub 0.0))
; Label this contact as the substrate/body.
  "substrate")

; Define uniform doping profiles for substrate and gate regions.
(sdedr:define-constant-profile "Const.Substrate"
  DopSub SubDop )
(sdedr:define-constant-profile-region "PlaceCD.Substrate" "Const.Substrate" "R.Substrate" )

(sdedr:define-constant-profile "Const.Gate"
  DopSD 1e20 )
(sdedr:define-constant-profile-region "PlaceCD.Gate" "Const.Gate" "R.Polygate" )

; Define a geometric window or baseline used for implant placement or mesh refinement.
(sdedr:define-refeval-window "BaseLine.Drain" "Line"
  (position (+ Xg 0.035) 0.0 0.0 )

```

```

(position (* Xmax 2.0) 0.0 0.0 ) )

; Define Gaussian implant profiles for source/drain, LDD extension, or halo regions.
(sdedr:define-gaussian-profile "Impl.SDprof"
DopSD
; Set a very high implant peak concentration.
"PeakPos" 0 "PeakVal" 1e20
"ValueAtDepth" SubDop "Depth" Xj
"Gauss" "Factor" 0.4
)

; Source/drain implant profile: high peak concentration for heavily doped drain/source.
(sdedr:define-analytical-profile-placement "Impl.Drain" "Impl.SDprof" "BaseLine.Drain" "Symm" "NoReplace" "Eval")


; Define mesh refinement sizes for important regions.
(sdedr:define-refinement-size "Ref.Substrate"
(/ Ltot 4.0) (/ Hsub 8.0)
Gpn Gpn )
; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-function "Ref.Substrate" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-region "RefPlace.Substrate" "Ref.Substrate" "R.Substrate" )

(sdedr:define-refeval-window "RWin.Act" "Rectangle"
(position 0.0 0.0 0.0)
(position Xmax (* Xj 1.2) 0.0 ) )
(sdedr:define-refinement-size "Ref.SiAct"
(/ Lcont 4.0) (/ Xj 8.0)
Gpn Gpn )

; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-function "Ref.SiAct" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-placement "RefPlace.SiAct" "Ref.SiAct" "RWin.Act" )

(sdedr:define-refinement-size "Ref.GOX"
(/ Ltot 4.0) (/ Tox 4.0)
Gpn (/ Tox 8.0) )
(sdedr:define-refinement-region "RefPlace.GOX" "Ref.GOX" "R.Gateox" )

(sdedr:define-refeval-window "MBWindow.Gate" "Rectangle"
(position (* Xg -1.0) Ypol 0.0)
(position (* Xg 1.0) Ygox 0.0))

; Use multibox refinement to create a fine mesh near the gate/channel and a coarser mesh away from it.
(sdedr:define-multibox-size "MBSIZE.Gate"
(/ Lg 8.0) (/ Hpol 8.0)
(/ Lg 10.0) 2e-4
1.0 -1.35)
(sdedr:define-multibox-placement "MBPlace.Gate" "MBSIZE.Gate" "MBWindow.Gate")

(sdedr:define-refeval-window "MBWindow.Channel" "Rectangle"
(position 0.0 0.0 0.0)
(position (* Xg 1.2) (* 0.5 Xj) 0.0) )
(sdedr:define-multibox-size "MBSIZE.Channel"
(/ Lg 8.0) (/ Xj 8.0)
(/ Lg 10.0) 1e-4
1.0 1.35)
(sdedr:define-multibox-placement "MBPlace.Channel" "MBSIZE.Channel" "MBWindow.Channel")

```

```
(sdedr:define-refinement-size "Ref.J"  
  (/ 0.016 6.0) 99  
  (/ 0.016 7.0) 66)  
  
(sdedr:define-refeval-window "RWin.GD" "Rectangle"  
  (position (- Xg 0.01) 0.0
```

2. LONG SDEVICE transfer_curve CODE

- Builds the long-channel reference MOSFET structure.
- Defines substrate/gate regions, source/drain doping, contacts, mesh refinement, and mirror symmetry.

Annotated Code

```
!(  
; Switch simulation settings between nMOS and pMOS cases.  
if { "@Type@" == "nMOS" } {  
; SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN 1.0  
; Select density-gradient quantum correction for electrons or holes.  
set DG "eQuantumPotential"  
; Select carrier temperature variable for hydrodynamic transport.  
set cTemp "eTemperature"  
; Set the gate work function used by the gate electrode.  
set WF 4.2  
; Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson eQuantumPotential Electron "  
; Define the full coupled equation set for the final device solution.  
set EQNS "Poisson eQuantumPotential Electron Hole eTemperature"  
} else {  
; SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN -1.0  
; Select density-gradient quantum correction for electrons or holes.  
set DG "hQuantumPotential"  
; Select carrier temperature variable for hydrodynamic transport.  
set cTemp "hTemperature"  
; Set the gate work function used by the gate electrode.  
set WF 4.8  
; Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson hQuantumPotential Hole"  
; Define the full coupled equation set for the final device solution.  
set EQNS "Poisson hQuantumPotential Electron Hole hTemperature"  
}  
)!  
  
; File block maps Workbench input/output files for mesh, parameters, plots, currents, and Logs.  
File {  
*input files:  
; Input mesh generated from SDE/SProcess.  
Grid= "@tdr@"  
; Input mechanical/strain file, if available.  
Piezo= "@tdr@"  
; Parameter file passed from Sentaurus Workbench.  
Parameter= "@parameter@"  
*output files:  
; TDR output file for spatial physical quantities.  
Plot= "@tdrdat@"  
; Output file for terminal currents.  
Current= "@plot@"  
; Log/output file for solver messages.  
Output= "@log@"  
}  
  
; Define terminal boundary conditions and contact properties.  
Electrode {  
; Source terminal is grounded.  
{ Name= "source" Voltage= 0.0 Resistor = 40 }  
; Drain terminal bias is initialized, then later ramped/swept.  
{ Name= "drain" Voltage= 0.0 Resistor = 40 }
```

```

; Gate terminal is initialized and assigned polysilicon/work-function behavior.
{ Name= "gate" Voltage= 0.0 Workfunction = !(puts $WF)! }
; Body/substrate terminal is grounded.
{ Name= "substrate" Voltage= 0.0}
}

; Enable global physical models used by SDevice.
Physics {
; Use Fermi-Dirac carrier statistics for highly doped regions.
Fermi
; Enable hydrodynamic transport to account for carrier heating/high-field effects.
Hydrodynamic( !(puts $cTemp)! )
}
#include "Plot_Math_des.cmd"

; Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
* =Creating initial guess:
NewCurrentFile="n@node@_init"
; Solve Poisson and transport equations self-consistently.
Coupled(Iterations= 1000 LineSearchDamping = 1e-4){ Poisson !(puts $DG)!}
Coupled { !(puts $EQN0)! }
Coupled { !(puts $EQNS)! }

*-ramp to gate to negative
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
; Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
; Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
; Gate terminal is initialized and assigned polysilicon/work-function behavior.
Goal { Name="gate" Voltage = -2.0}
; Solve Poisson and transport equations self-consistently.
) { Coupled { Poisson !(puts $DG)! }}

*-ramp to drain to Vd
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
; Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
; Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
; Drain terminal bias is initialized, then later ramped/swept.
Goal { Name="drain" Voltage = !(puts [expr $SIGN*@Vd@])!}
){ Coupled { !(puts $EQNS)! }}

; Save spatial solution snapshots for visualization.
Plot ( FilePrefix = "n@node@_GIDL_" )

*-Vg sweep
; Set current-output file prefix for this sweep.
NewCurrentPrefix=""
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
doZero
; Control numerical step size for convergence and resolution.
InitialStep= 1e-3 Increment = 1.5
; Control numerical step size for convergence and resolution.
MinStep = 1e-7 MaxStep = 0.04
; Bias target for the current ramp or sweep.
Goal { Name = "gate" Voltage = !(puts [expr $SIGN*@Vg@])!}
){ Coupled { !(puts $EQNS)! }
; Sample terminal currents during the sweep.
CurrentPlot( Time=(Range=(0 1) Intervals = 30) )
}

```


3. LONG SDEVICE output_CURVE CODE

- Builds the long-channel reference MOSFET structure.
- Defines substrate/gate regions, source/drain doping, contacts, mesh refinement, and mirror symmetry.

Annotated Code

```
!(  
; Switch simulation settings between nMOS and pMOS cases.  
if { "@Type@" == "nMOS" } {  
; SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN 1.0  
; Select density-gradient quantum correction for electrons or holes.  
set DG "eQuantumPotential"  
; Select carrier temperature variable for hydrodynamic transport.  
set cTemp "eTemperature"  
; Set the gate work function used by the gate electrode.  
set WF 4.2  
; Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson eQuantumPotential Electron "  
; Define the full coupled equation set for the final device solution.  
set EQNS "Poisson eQuantumPotential Electron Hole eTemperature"  
} else {  
; SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN -1.0  
; Select density-gradient quantum correction for electrons or holes.  
set DG "hQuantumPotential"  
; Select carrier temperature variable for hydrodynamic transport.  
set cTemp "hTemperature"  
; Set the gate work function used by the gate electrode.  
set WF 4.8  
; Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson hQuantumPotential Hole"  
; Define the full coupled equation set for the final device solution.  
set EQNS "Poisson hQuantumPotential Electron Hole hTemperature"  
}  
)!  
  
; File block maps Workbench input/output files for mesh, parameters, plots, currents, and Logs.  
File {  
*input files:  
; Input mesh generated from SDE/SProcess.  
Grid= "@tdr@"  
; Input mechanical/strain file, if available.  
Piezo= "@tdr@"  
; Parameter file passed from Sentaurus Workbench.  
Parameter= "@parameter@"  
*output files:  
; TDR output file for spatial physical quantities.  
Plot= "@tdrdat@"  
; Output file for terminal currents.  
Current= "@plot@"  
; Log/output file for solver messages.  
Output= "@log@"  
}  
  
; Define terminal boundary conditions and contact properties.  
Electrode {  
; Source terminal is grounded.  
{ Name= "source" Voltage= 0.0 Resistor = 40 }  
; Drain terminal bias is initialized, then later ramped/swept.  
{ Name= "drain" Voltage= 0.0 Resistor = 40 }
```

```

; Gate terminal is initialized and assigned polysilicon/work-function behavior.
{ Name= "gate" Voltage= 0.0 Workfunction = !(puts $WF)! }
; Body/substrate terminal is grounded.
{ Name= "substrate" Voltage= 0.0}
}

; Enable global physical models used by SDevice.
Physics {
; Use Fermi-Dirac carrier statistics for highly doped regions.
Fermi
; Enable hydrodynamic transport to account for carrier heating/high-field effects.
Hydrodynamic( !(puts $cTemp)! )
}
#include "Plot_Math_des.cmd"

; Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
* =Creating initial guess:
NewCurrentFile="n@node@_init"
; Solve Poisson and transport equations self-consistently.
Coupled(Iterations= 1000 LineSearchDamping = 1e-4){ Poisson !(puts $DG)!}
Coupled { !(puts $EQN0)! }
Coupled { !(puts $EQNS)! }

*-ramp to drain to Vd
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
; Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
; Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
; Gate terminal is initialized and assigned polysilicon/work-function behavior.
Goal { Name="gate" Voltage = !(puts [expr $SIGN*@Vgg@])!}
){ Coupled { !(puts $EQNS)! }}

; Save spatial solution snapshots for visualization.
Plot ( FilePrefix = "n@node@@Vgg@Vg_" )

*-Vg sweep
; Set current-output file prefix for this sweep.
NewCurrentPrefix="IdVd_"
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
doZero
; Control numerical step size for convergence and resolution.
InitialStep= 1e-3 Increment = 1.5
; Control numerical step size for convergence and resolution.
MinStep = 1e-7 MaxStep = 0.04
; Bias target for the current ramp or sweep.
Goal { Name = "drain" Voltage = !(puts [expr $SIGN*@Vdd@])!}
){ Coupled { !(puts $EQNS)! }
; Sample terminal currents during the sweep.
CurrentPlot( Time=(Range=(0 1) Intervals = 30) )
}

```

4. LONG GIDL SDE CODE

- Builds a MOSFET structure for GIDL analysis.
- Uses a sharper drain implant profile to increase the high-field region near the drain edge.

Annotated Code

```
; Clear any previously defined SDE geometry, doping, mesh, and contacts.
(sde:clear)
; Define the process-up direction so vertical coordinates are interpreted consistently.
(sde:set-process-up-direction "+z")

; Select doping species and carrier models based on whether the device is nMOS or pMOS.
(if (string=? "@Type@" "nMOS")
    (begin
; Define the substrate dopant species.
      (define DopSub "BoronActiveConcentration")
; Define the source/drain and gate dopant species.
      (define DopSD "ArsenicActiveConcentration")
; Set the background substrate doping concentration.
      (define SubDop 1e18 )
    )
    (begin
; Define the substrate dopant species.
      (define DopSub "ArsenicActiveConcentration")
; Define the source/drain and gate dopant species.
      (define DopSD "BoronActiveConcentration")
; Set the background substrate doping concentration.
      (define SubDop 1e18)
    )
)

; Gate Length parameter imported from Sentaurus Workbench.
(define Lg @lgate@)
; Total simulated lateral length, including gate and source/drain extension region.
(define Ltot (+ Lg (* 2 0.01) 0.8) )

; Substrate thickness used for the 2D device cross-section.
(define Hsub 1.0)
; Gate oxide thickness parameter imported from Workbench.
(define Tox @Toxide@)
; Polysilicon gate height.
(define Hpol 0.2)

; Minimum mesh size control used near critical regions.
(define Gpn 0.005)

; Approximate junction depth used for source/drain implants.
(define Xj 0.1 )

; Derived dimensions are computed from the primary process/device parameters.
;Derived quantities
; Half-device lateral boundary for building a symmetric structure.
(define Xmax (/ Ltot 2.0))
; Half gate length.
(define Xg (/ Lg 2.0))

; Bottom substrate coordinate.
(define Ysub Hsub)
; Gate oxide top/bottom coordinate based on oxide thickness.
(define Ygox (* Tox -1.0))
```

```

; Top polysilicon coordinate.
(define Ypol (- Ygox Hpol))

(define Lcont Xmax)

; Set the default Boolean operation for overlapping SDE geometry creation.
(sdegeo:set-default-boolean "ABA")

; Create a 2D rectangular material region such as silicon, oxide, spacer, or polysilicon.
(sdegeo:create-rectangle
  (position 0.0 Ysub 0.0 )
  (position Xmax 0.0 0.0 )
; Name the silicon body/substrate region.
  "Silicon" "R.Substrate"
)

(sdegeo:create-rectangle
  (position 0.0 0.0 0.0 )
  (position Xg Ygox 0.0 )
; Name the gate oxide region.
  "Oxide" "R.Gateox"
)

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xg Ypol 0.0 )
; Define the re-oxidized oxide region around the polysilicon gate.
  "Oxide" "R.PolyReox"
)

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xg Ypol 0.0 )
; Define the polysilicon gate region.
  "PolySi" "R.Polygate"
)

; Assign electrical contacts to selected geometry edges.
(sdegeo:set-contact
  (find-edge-id (position (* (+ Xmax Xg) 0.5) 0.0 0.0))
; Label this contact as the drain.
  "drain")
(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ypol 0.0))
; Label this contact as the gate.
  "gate")
(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ysub 0.0))
; Label this contact as the substrate/body.
  "substrate")

; Define uniform doping profiles for substrate and gate regions.
(sdedr:define-constant-profile "Const.Substrate"
  DopSub SubDop )
(sdedr:define-constant-profile-region "PlaceCD.Substrate" "Const.Substrate" "R.Substrate" )

(sdedr:define-constant-profile "Const.Gate"
  DopSD 1e20 )
(sdedr:define-constant-profile-region "PlaceCD.Gate" "Const.Gate" "R.Polygate" )

; Define a geometric window or baseline used for implant placement or mesh refinement.
(sdedr:define-refeval-window "BaseLine.Drain" "Line"
  (position (+ Xg 0.035) 0.0 0.0 )

```

```

(position (* Xmax 2.0) 0.0 0.0 ) )

; Define Gaussian implant profiles for source/drain, LDD extension, or halo regions.
(sdedr:define-gaussian-profile "Impl.SDprof"
DopSD
; Set a very high implant peak concentration.
"PeakPos" 0 "PeakVal" 1e20
"ValueAtDepth" SubDop "Depth" Xj
"Gauss" "Factor" 0.9
)

; Source/drain implant profile: high peak concentration for heavily doped drain/source.
(sdedr:define-analytical-profile-placement "Impl.Drain" "Impl.SDprof" "BaseLine.Drain" "Symm" "NoReplace" "Eval")


; Define mesh refinement sizes for important regions.
(sdedr:define-refinement-size "Ref.Substrate"
(/ Ltot 4.0) (/ Hsub 8.0)
Gpn Gpn )
; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-function "Ref.Substrate" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-region "RefPlace.Substrate" "Ref.Substrate" "R.Substrate" )

(sdedr:define-refeval-window "RWin.Act" "Rectangle"
(position 0.0 0.0 0.0)
(position Xmax (* Xj 1.2) 0.0 ) )
(sdedr:define-refinement-size "Ref.SiAct"
(/ Lcont 4.0) (/ Xj 8.0)
Gpn Gpn )

; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-function "Ref.SiAct" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-placement "RefPlace.SiAct" "Ref.SiAct" "RWin.Act" )

(sdedr:define-refinement-size "Ref.GOX"
(/ Ltot 4.0) (/ Tox 4.0)
Gpn (/ Tox 8.0) )
(sdedr:define-refinement-region "RefPlace.GOX" "Ref.GOX" "R.Gateox" )

(sdedr:define-refeval-window "MBWindow.Gate" "Rectangle"
(position (* Xg -1.0) Ypol 0.0)
(position (* Xg 1.0) Ygox 0.0))

; Use multibox refinement to create a fine mesh near the gate/channel and a coarser mesh away from it.
(sdedr:define-multibox-size "MBSIZE.Gate"
(/ Lg 8.0) (/ Hpol 8.0)
(/ Lg 10.0) 2e-4
1.0 -1.35)
(sdedr:define-multibox-placement "MBPlace.Gate" "MBSIZE.Gate" "MBWindow.Gate")


(sdedr:define-refeval-window "MBWindow.Channel" "Rectangle"
(position 0.0 0.0 0.0)
(position (* Xg 1.2) (* 0.5 Xj) 0.0) )
(sdedr:define-multibox-size "MBSIZE.Channel"
(/ Lg 8.0) (/ Xj 8.0)
(/ Lg 10.0) 1e-4
1.0 1.35)
(sdedr:define-multibox-placement "MBPlace.Channel" "MBSIZE.Channel" "MBWindow.Channel")

```

```
(sdedr:define-refinement-size "Ref.J"  
  (/ 0.016 6.0) 99  
  (/ 0.016 7.0) 66)  
  
(sdedr:define-refeval-window "RWin.GD" "Rectangle"  
  (position (- Xg 0.01) 0.0 0.0)  
  (position (+ Xg 0) (* Xj 0.2) 0.0) )  
(sdedr:define-refinement-placement "RefPlace.GJ" "Ref.J" "RWin.GD" )
```

```
; Generate the half-device mesh file.
```

```
(sde:build-mesh "n@node@_half_msh")
```

```
; Reflect the half-device structure to build the full symmetric MOSFET and rename the mirrored contact as source.
```

```
(system:command "tdx -mtt -x -M 0 -S 0 -ren drain=source n@node@_half_msh n@node@_msh")
```

5. LONG GIDL SDEVICE TRANSFERCURVE CODE

- Builds a MOSFET structure for GIDL analysis.
- Uses a sharper drain implant profile to increase the high-field region near the drain edge.

Annotated Code

```
!(  
; Switch simulation settings between nMOS and pMOS cases.  
if { "@Type@" == "nMOS" } {  
; SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN 1.0  
; Select density-gradient quantum correction for electrons or holes.  
set DG "eQuantumPotential"  
; Select carrier temperature variable for hydrodynamic transport.  
set cTemp "eTemperature"  
; Set the gate work function used by the gate electrode.  
set WF 4.2  
; Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson eQuantumPotential Electron "  
; Define the full coupled equation set for the final device solution.  
set EQNS "Poisson eQuantumPotential Electron Hole eTemperature"  
} else {  
; SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN -1.0  
; Select density-gradient quantum correction for electrons or holes.  
set DG "hQuantumPotential"  
; Select carrier temperature variable for hydrodynamic transport.  
set cTemp "hTemperature"  
; Set the gate work function used by the gate electrode.  
set WF 4.8  
; Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson hQuantumPotential Hole"  
; Define the full coupled equation set for the final device solution.  
set EQNS "Poisson hQuantumPotential Electron Hole hTemperature"  
}  
)!  
  
; File block maps Workbench input/output files for mesh, parameters, plots, currents, and Logs.  
File {  
*input files:  
; Input mesh generated from SDE/SProcess.  
Grid= "@tdr@"  
; Input mechanical/strain file, if available.  
Piezo= "@tdr@"  
; Parameter file passed from Sentaurus Workbench.  
Parameter= "@parameter@"  
*output files:  
; TDR output file for spatial physical quantities.  
Plot= "@tdrdat@"  
; Output file for terminal currents.  
Current= "@plot@"  
; Log/output file for solver messages.  
Output= "@log@"  
}  
  
; Define terminal boundary conditions and contact properties.  
Electrode {  
; Source terminal is grounded.  
{ Name= "source" Voltage= 0.0 Resistor = 40 }  
; Drain terminal bias is initialized, then later ramped/swept.  
{ Name= "drain" Voltage= 0.0 Resistor = 40 }
```

```

; Gate terminal is initialized and assigned polysilicon/work-function behavior.
{ Name= "gate" Voltage= 0.0 Workfunction = !(puts $WF)! }
; Body/substrate terminal is grounded.
{ Name= "substrate" Voltage= 0.0}
}

; Enable global physical models used by SDevice.
Physics {
; Use Fermi-Dirac carrier statistics for highly doped regions.
Fermi
; Enable hydrodynamic transport to account for carrier heating/high-field effects.
Hydrodynamic( !(puts $cTemp)! )

Recombination (
; Enable nonlocal band-to-band tunneling model for GIDL simulation.
Band2Band(Model=NonlocalPath)
)
}
#include "Plot_Math_des.cmd"

; Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
* =Creating initial guess:
NewCurrentFile="n@node@_init"
; Solve Poisson and transport equations self-consistently.
Coupled(Iterations= 1000 LineSearchDamping = 1e-4){ Poisson !(puts $DG)!}
Coupled { !(puts $EQN0)! }
Coupled { !(puts $EQNS)! }

*-ramp to gate to negative
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
; Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
; Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
; Gate terminal is initialized and assigned polysilicon/work-function behavior.
Goal { Name="gate" Voltage = -2.0}
; Solve Poisson and transport equations self-consistently.
) { Coupled { Poisson !(puts $DG)! }}

*-ramp to drain to Vd
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
; Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
; Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
; Drain terminal bias is initialized, then later ramped/swept.
Goal { Name="drain" Voltage = !(puts [expr $SIGN*@Vd@])!}
){ Coupled { !(puts $EQNS)! }}

; Save spatial solution snapshots for visualization.
Plot ( FilePrefix = "n@node@_GIDL_" )

*-Vg sweep
; Set current-output file prefix for this sweep.
NewCurrentPrefix=""
; Perform a DC bias ramp with adaptive step control.
Quasistationary(
doZero
; Control numerical step size for convergence and resolution.
InitialStep= 1e-3 Increment = 1.5
; Control numerical step size for convergence and resolution.
MinStep = 1e-7 MaxStep = 0.04
; Bias target for the current ramp or sweep.

```

```
Goal { Name = "gate" Voltage = !(puts [expr $SIGN*@Vg@])!}  
) { Coupled { !(puts $EQNS)! }  
; Sample terminal currents during the sweep.  
CurrentPlot( Time=(Range=(0 1) Intervals = 30) )  
}  
}
```

6. LONG GIDL OUTPUTCURVE CODE

- Runs an output-curve sweep for the GIDL structure.
- Used to compare current response under fixed gate bias and varying drain bias.

Annotated Code

```
!(  
* Switch simulation settings between nMOS and pMOS cases.  
if { "@Type@" == "nMOS" } {  
* SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN 1.0  
* Select density-gradient quantum correction for electrons or holes.  
set DG "eQuantumPotential"  
* Select carrier temperature variable for hydrodynamic transport.  
set cTemp "eTemperature"  
* Set the gate work function used by the gate electrode.  
set WF 4.2  
* Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson eQuantumPotential Electron "  
* Define the full coupled equation set for the final device solution.  
set EQNS "Poisson eQuantumPotential Electron Hole eTemperature"  
} else {  
* SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN -1.0  
* Select density-gradient quantum correction for electrons or holes.  
set DG "hQuantumPotential"  
* Select carrier temperature variable for hydrodynamic transport.  
set cTemp "hTemperature"  
* Set the gate work function used by the gate electrode.  
set WF 4.8  
* Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson hQuantumPotential Hole"  
* Define the full coupled equation set for the final device solution.  
set EQNS "Poisson hQuantumPotential Electron Hole hTemperature"  
}  
)!  
  
* File block maps Workbench input/output files for mesh, parameters, plots, currents, and Logs.  
File {  
*input files:  
* Input mesh generated from SDE/SProcess.  
Grid= "@tdr@"  
* Input mechanical/strain file, if available.  
Piezo= "@tdr@"  
* Parameter file passed from Sentaurus Workbench.  
Parameter= "@parameter@"  
*output files:  
* TDR output file for spatial physical quantities.  
Plot= "@tdrdat@"  
* Output file for terminal currents.  
Current= "@plot@"  
* Log/output file for solver messages.  
Output= "@log@"  
}  
  
* Define terminal boundary conditions and contact properties.  
Electrode {  
* Source terminal is grounded.  
{ Name= "source" Voltage= 0.0 Resistor = 40 }  
* Drain terminal bias is initialized, then later ramped/swept.  
{ Name= "drain" Voltage= 0.0 Resistor = 40 }
```

```

* Gate terminal is initialized and assigned polysilicon/work-function behavior.
{ Name= "gate" Voltage= 0.0 Workfunction = !(puts $WF)! }
* Body/substrate terminal is grounded.
{ Name= "substrate" Voltage= 0.0}
}

* Enable global physical models used by SDevice.
Physics {
* Use Fermi-Dirac carrier statistics for highly doped regions.
Fermi
* Enable hydrodynamic transport to account for carrier heating/high-field effects.
Hydrodynamic( !(puts $cTemp)! )
}
#include "Plot_Math_des.cmd"

* Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
* =Creating initial guess:
NewCurrentFile="n@node@_init"
* Solve Poisson and transport equations self-consistently.
Coupled(Iterations= 1000 LineSearchDamping = 1e-4){ Poisson !(puts $DG)!}
Coupled { !(puts $EQN0)! }
Coupled { !(puts $EQNS)! }

*-ramp to drain to Vd
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
* Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
* Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
* Gate terminal is initialized and assigned polysilicon/work-function behavior.
Goal { Name="gate" Voltage = !(puts [expr $SIGN*@Vgg@])!}
){ Coupled { !(puts $EQNS)! }}

* Save spatial solution snapshots for visualization.
Plot ( FilePrefix = "n@node@@Vgg@Vg_" )

*-Vg sweep
* Set current-output file prefix for this sweep.
NewCurrentPrefix="IdVd_"
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
doZero
* Control numerical step size for convergence and resolution.
InitialStep= 1e-3 Increment = 1.5
* Control numerical step size for convergence and resolution.
MinStep = 1e-7 MaxStep = 0.04
* Bias target for the current ramp or sweep.
Goal { Name = "drain" Voltage = !(puts [expr $SIGN*@Vdd@])!}
){ Coupled { !(puts $EQNS)! }
* Sample terminal currents during the sweep.
CurrentPlot( Time=(Range=(0 1) Intervals = 30) )
}
}

```

7. SHORT SDE CODE

- Builds the short-channel MOSFET structure.
- This structure is used to observe short-channel effects such as Vth roll-off, DIBL, and leakage increase.

Annotated Code

```
; Clear any previously defined SDE geometry, doping, mesh, and contacts.
(sde:clear)
; Define the process-up direction so vertical coordinates are interpreted consistently.
(sde:set-process-up-direction "+z")

; Select doping species and carrier models based on whether the device is nMOS or pMOS.
(if (string=? "@Type@" "nMOS")
    (begin
; Define the substrate dopant species.
      (define DopSub "BoronActiveConcentration")
; Define the source/drain and gate dopant species.
      (define DopSD "ArsenicActiveConcentration")
; Set the background substrate doping concentration.
      (define SubDop 1e18 )
    )
    (begin
; Define the substrate dopant species.
      (define DopSub "ArsenicActiveConcentration")
; Define the source/drain and gate dopant species.
      (define DopSD "BoronActiveConcentration")
; Set the background substrate doping concentration.
      (define SubDop 1e18)
    )
)

; Gate Length parameter imported from Sentaurus Workbench.
(define Lg @lgate@)
; Total simulated lateral length, including gate and source/drain extension region.
(define Ltot (+ Lg (* 2 0.01) 0.8) )

; Substrate thickness used for the 2D device cross-section.
(define Hsub 1.0)
; Gate oxide thickness parameter imported from Workbench.
(define Tox @Toxide@)
; Polysilicon gate height.
(define Hpol 0.2)

; Minimum mesh size control used near critical regions.
(define Gpn 0.005)

; Approximate junction depth used for source/drain implants.
(define Xj 0.1 )

; Derived dimensions are computed from the primary process/device parameters.
;Derived quantities
; Half-device lateral boundary for building a symmetric structure.
(define Xmax (/ Ltot 2.0))
; Half gate length.
(define Xg (/ Lg 2.0))

; Bottom substrate coordinate.
(define Ysub Hsub)
; Gate oxide top/bottom coordinate based on oxide thickness.
(define Ygox (* Tox -1.0))
```

```

; Top polysilicon coordinate.
(define Ypol (- Ygox Hpol))

(define Lcont Xmax)

; Set the default Boolean operation for overlapping SDE geometry creation.
(sdegeo:set-default-boolean "ABA")

; Create a 2D rectangular material region such as silicon, oxide, spacer, or polysilicon.
(sdegeo:create-rectangle
  (position 0.0 Ysub 0.0 )
  (position Xmax 0.0 0.0 )
; Name the silicon body/substrate region.
  "Silicon" "R.Substrate"
)

(sdegeo:create-rectangle
  (position 0.0 0.0 0.0 )
  (position Xg Ygox 0.0 )
; Name the gate oxide region.
  "Oxide" "R.Gateox"
)

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xg Ypol 0.0 )
; Define the re-oxidized oxide region around the polysilicon gate.
  "Oxide" "R.PolyReox"
)

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xg Ypol 0.0 )
; Define the polysilicon gate region.
  "PolySi" "R.Polygate"
)

; Assign electrical contacts to selected geometry edges.
(sdegeo:set-contact
  (find-edge-id (position (* (+ Xmax Xg) 0.5) 0.0 0.0))
; Label this contact as the drain.
  "drain")
(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ypol 0.0))
; Label this contact as the gate.
  "gate")
(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ysub 0.0))
; Label this contact as the substrate/body.
  "substrate")

; Define uniform doping profiles for substrate and gate regions.
(sdedr:define-constant-profile "Const.Substrate"
  DopSub SubDop )
(sdedr:define-constant-profile-region "PlaceCD.Substrate" "Const.Substrate" "R.Substrate" )

(sdedr:define-constant-profile "Const.Gate"
  DopSD 1e20 )
(sdedr:define-constant-profile-region "PlaceCD.Gate" "Const.Gate" "R.Polygate" )

; Define a geometric window or baseline used for implant placement or mesh refinement.
(sdedr:define-refeval-window "BaseLine.Drain" "Line"
  (position (+ Xg 0.035) 0.0 0.0 )

```

```

(position (* Xmax 2.0) 0.0 0.0 ) )

; Define Gaussian implant profiles for source/drain, LDD extension, or halo regions.
(sdedr:define-gaussian-profile "Impl.SDprof"
DopSD
; Set a very high implant peak concentration.
"PeakPos" 0 "PeakVal" 1e20
"ValueAtDepth" SubDop "Depth" Xj
"Gauss" "Factor" 0.4
)

; Source/drain implant profile: high peak concentration for heavily doped drain/source.
(sdedr:define-analytical-profile-placement "Impl.Drain" "Impl.SDprof" "BaseLine.Drain" "Symm" "NoReplace" "Eval")


; Define mesh refinement sizes for important regions.
(sdedr:define-refinement-size "Ref.Substrate"
(/ Ltot 4.0) (/ Hsub 8.0)
Gpn Gpn )
; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-function "Ref.Substrate" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-region "RefPlace.Substrate" "Ref.Substrate" "R.Substrate" )

(sdedr:define-refeval-window "RWin.Act" "Rectangle"
(position 0.0 0.0 0.0)
(position Xmax (* Xj 1.2) 0.0 ) )
(sdedr:define-refinement-size "Ref.SiAct"
(/ Lcont 4.0) (/ Xj 8.0)
Gpn Gpn )

; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-function "Ref.SiAct" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-placement "RefPlace.SiAct" "Ref.SiAct" "RWin.Act" )

(sdedr:define-refinement-size "Ref.GOX"
(/ Ltot 4.0) (/ Tox 4.0)
Gpn (/ Tox 8.0) )
(sdedr:define-refinement-region "RefPlace.GOX" "Ref.GOX" "R.Gateox" )

(sdedr:define-refeval-window "MBWindow.Gate" "Rectangle"
(position (* Xg -1.0) Ypol 0.0)
(position (* Xg 1.0) Ygox 0.0))

; Use multibox refinement to create a fine mesh near the gate/channel and a coarser mesh away from it.
(sdedr:define-multibox-size "MBSIZE.Gate"
(/ Lg 8.0) (/ Hpol 8.0)
(/ Lg 10.0) 2e-4
1.0 -1.35)
(sdedr:define-multibox-placement "MBPlace.Gate" "MBSIZE.Gate" "MBWindow.Gate")


(sdedr:define-refeval-window "MBWindow.Channel" "Rectangle"
(position 0.0 0.0 0.0)
(position (* Xg 1.2) (* 0.5 Xj) 0.0) )
(sdedr:define-multibox-size "MBSIZE.Channel"
(/ Lg 8.0) (/ Xj 8.0)
(/ Lg 10.0) 1e-4
1.0 1.35)
(sdedr:define-multibox-placement "MBPlace.Channel" "MBSIZE.Channel" "MBWindow.Channel")

```

```
(sdedr:define-refinement-size "Ref.J"  
  (/ 0.016 6.0) 99  
  (/ 0.016 7.0) 66)  
  
(sdedr:define-refeval-window "RWin.GD" "Rectangle"  
  (position (- Xg 0.01) 0.0 0.0)  
  (position (+ Xg 0) (* Xj 0.2) 0.0) )  
(sdedr:define-refinement-placement "RefPlace.GJ" "Ref.J" "RWin.GD" )
```

```
; Generate the half-device mesh file.
```

```
(sde:build-mesh "n@node@_half_msh")
```

```
; Reflect the half-device structure to build the full symmetric MOSFET and rename the mirrored contact as source.
```

```
(system:command "tdx -mtt -x -M 0 -S 0 -ren drain=source n@node@_half_msh n@node@_msh")
```

8. SHORT TRANSFER CODE

- Runs the short-channel Id-Vg transfer simulation.
- The result is compared with long-channel behavior to analyze subthreshold degradation.

Annotated Code

```
!(  
* Switch simulation settings between nMOS and pMOS cases.  
if { "@Type@" == "nMOS" } {  
* SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN 1.0  
* Select density-gradient quantum correction for electrons or holes.  
set DG "eQuantumPotential"  
* Select carrier temperature variable for hydrodynamic transport.  
set cTemp "eTemperature"  
* Set the gate work function used by the gate electrode.  
set WF 4.2  
* Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson eQuantumPotential Electron "  
* Define the full coupled equation set for the final device solution.  
set EQNS "Poisson eQuantumPotential Electron Hole eTemperature"  
} else {  
* SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.  
set SIGN -1.0  
* Select density-gradient quantum correction for electrons or holes.  
set DG "hQuantumPotential"  
* Select carrier temperature variable for hydrodynamic transport.  
set cTemp "hTemperature"  
* Set the gate work function used by the gate electrode.  
set WF 4.8  
* Define the reduced equation set used after the initial Poisson solve.  
set EQN0 "Poisson hQuantumPotential Hole"  
* Define the full coupled equation set for the final device solution.  
set EQNS "Poisson hQuantumPotential Electron Hole hTemperature"  
}  
)!  
  
* File block maps Workbench input/output files for mesh, parameters, plots, currents, and Logs.  
File {  
*input files:  
* Input mesh generated from SDE/SProcess.  
Grid= "@tdr@"  
* Input mechanical/strain file, if available.  
Piezo= "@tdr@"  
* Parameter file passed from Sentaurus Workbench.  
Parameter= "@parameter@"  
*output files:  
* TDR output file for spatial physical quantities.  
Plot= "@tdrdat@"  
* Output file for terminal currents.  
Current= "@plot@"  
* Log/output file for solver messages.  
Output= "@log@"  
}  
  
* Define terminal boundary conditions and contact properties.  
Electrode {  
* Source terminal is grounded.  
{ Name= "source" Voltage= 0.0 Resistor = 40 }  
* Drain terminal bias is initialized, then later ramped/swept.  
{ Name= "drain" Voltage= 0.0 Resistor = 40 }
```

```

* Gate terminal is initialized and assigned polysilicon/work-function behavior.
{ Name= "gate" Voltage= 0.0 Workfunction = !(puts $WF)! }
* Body/substrate terminal is grounded.
{ Name= "substrate" Voltage= 0.0}
}

* Enable global physical models used by SDevice.
Physics {
* Use Fermi-Dirac carrier statistics for highly doped regions.
Fermi
* Enable hydrodynamic transport to account for carrier heating/high-field effects.
Hydrodynamic( !(puts $cTemp)! )
}
#include "Plot_Math_des.cmd"

* Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
* =Creating initial guess:
NewCurrentFile="n@node@_init"
* Solve Poisson and transport equations self-consistently.
Coupled(Iterations= 1000 LineSearchDamping = 1e-4){ Poisson !(puts $DG)!}
Coupled { !(puts $EQN0)! }
Coupled { !(puts $EQNS)! }

*-ramp to gate to negative
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
* Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
* Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
* Gate terminal is initialized and assigned polysilicon/work-function behavior.
Goal { Name="gate" Voltage = -2.0}
* Solve Poisson and transport equations self-consistently.
) { Coupled { Poisson !(puts $DG)! }}

*-ramp to drain to Vd
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
* Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
* Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
* Drain terminal bias is initialized, then later ramped/swept.
Goal { Name="drain" Voltage = !(puts [expr $SIGN*@Vd@])!}
){ Coupled { !(puts $EQNS)! }}

* Save spatial solution snapshots for visualization.
Plot ( FilePrefix = "n@node@_GIDL_" )

*-Vg sweep
* Set current-output file prefix for this sweep.
NewCurrentPrefix=""
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
doZero
* Control numerical step size for convergence and resolution.
InitialStep= 1e-3 Increment = 1.5
* Control numerical step size for convergence and resolution.
MinStep = 1e-7 MaxStep = 0.04
* Bias target for the current ramp or sweep.
Goal { Name = "gate" Voltage = !(puts [expr $SIGN*@Vg@])!}
){ Coupled { !(puts $EQNS)! }
* Sample terminal currents during the sweep.
CurrentPlot( Time=(Range=(0 1) Intervals = 30) )
}

```


9. Device Setup & Physical Models

- Annotated explanation of the repeated SDevice control block.
- This block selects nMOS/pMOS physics settings and performs a staged bias sweep.

Annotated Code

```
* Switch simulation settings between nMOS and pMOS cases.
if { "@Type@" == "nMOS" } {
* SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.
set SIGN 1.0
* Select density-gradient quantum correction for electrons or holes.
set DG "eQuantumPotential"
* Select carrier temperature variable for hydrodynamic transport.
set cTemp "eTemperature"
* Set the gate work function used by the gate electrode.
set WF 4.2
* Define the reduced equation set used after the initial Poisson solve.
set EQN0 "Poisson eQuantumPotential Electron "
* Define the full coupled equation set for the final device solution.
set EQNS "Poisson eQuantumPotential Electron Hole eTemperature"
} else {
* SIGN controls voltage polarity so the same template can run nMOS and pMOS simulations.
set SIGN -1.0
* Select density-gradient quantum correction for electrons or holes.
set DG "hQuantumPotential"
* Select carrier temperature variable for hydrodynamic transport.
set cTemp "hTemperature"
* Set the gate work function used by the gate electrode.
set WF 4.8
* Define the reduced equation set used after the initial Poisson solve.
set EQN0 "Poisson hQuantumPotential Hole"
* Define the full coupled equation set for the final device solution.
set EQNS "Poisson hQuantumPotential Electron Hole hTemperature"
}
}!

* File block maps Workbench input/output files for mesh, parameters, plots, currents, and logs.
File {
*input files:
* Input mesh generated from SDE/SProcess.
Grid= "@tldr@"
* Input mechanical/strain file, if available.
Piezo= "@tldr@"
* Parameter file passed from Sentaurus Workbench.
Parameter= "@parameter@"
*output files:
* TDR output file for spatial physical quantities.
Plot= "@tdrdat@"
* Output file for terminal currents.
Current= "@plot@"
* Log/output file for solver messages.
Output= "@log@"
}

* Define terminal boundary conditions and contact properties.
Electrode {
* Source terminal is grounded.
{ Name= "source" Voltage= 0.0 Resistor = 40 }
* Drain terminal bias is initialized, then later ramped/swept.
```

```

{ Name= "drain" Voltage= 0.0 Resistor = 40 }
* Gate terminal is initialized and assigned polysilicon/work-function behavior.
{ Name= "gate" Voltage= 0.0 Workfunction = !(puts $WF)! }
* Body/substrate terminal is grounded.
{ Name= "substrate" Voltage= 0.0}
}

* Enable global physical models used by SDevice.
Physics {
* Use Fermi-Dirac carrier statistics for highly doped regions.
Fermi
* Enable hydrodynamic transport to account for carrier heating/high-field effects.
Hydrodynamic( !(puts $cTemp)! )
}
#include "Plot_Math_des.cmd"

* Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
* =Creating initial guess:
NewCurrentFile="n@node@_init"
* Solve Poisson and transport equations self-consistently.
Coupled(Iterations= 1000 LineSearchDamping = 1e-4){ Poisson !(puts $DG)!}
Coupled { !(puts $EQN0)! }
Coupled { !(puts $EQNS)! }

*-ramp to gate to negative
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
* Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
* Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
* Gate terminal is initialized and assigned polysilicon/work-function behavior.
Goal { Name="gate" Voltage = @Vg@}
* Solve Poisson and transport equations self-consistently.
) { Coupled { Poisson !(puts $DG)! }}

*-ramp to drain to Vd
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
* Control numerical step size for convergence and resolution.
InitialStep= 1e-2 Increment = 1.35
* Control numerical step size for convergence and resolution.
MinStep= 1e-7 MaxStep = 0.2
* Drain terminal bias is initialized, then later ramped/swept.
Goal { Name="drain" Voltage = !(puts [expr $SIGN*@Vd@])!}
){ Coupled { !(puts $EQNS)! }}

* Save spatial solution snapshots for visualization.
Plot ( FilePrefix = "n@node@_GIDL_" )

*-Vg sweep
* Set current-output file prefix for this sweep.
NewCurrentPrefix=""
* Perform a DC bias ramp with adaptive step control.
Quasistationary(
doZero
* Control numerical step size for convergence and resolution.
InitialStep= 1e-3 Increment = 1.5
* Control numerical step size for convergence and resolution.
MinStep = 1e-7 MaxStep = 0.04
* Bias target for the current ramp or sweep.
Goal { Name = "gate" Voltage = !(puts [expr $SIGN*@Vg@])!}
){ Coupled { !(puts $EQNS)! }
* Sample terminal currents during the sweep.
CurrentPlot( Time=(Range=(0 1) Intervals = 30) )

```


10. LDD+HALO SDE CODE

- Builds an LDD + Halo MOSFET structure.
- Adds nitride spacer, lightly doped drain extension, and halo implant to suppress short-channel effects.

Annotated Code

```
; Clear any previously defined SDE geometry, doping, mesh, and contacts.
(sde:clear)
; Define the process-up direction so vertical coordinates are interpreted consistently.
(sde:set-process-up-direction "+z")

; Select doping species and carrier models based on whether the device is nMOS or pMOS.
(if (string=? "@Type@" "nMOS")
    (begin
; Define the substrate dopant species.
      (define DopSub "BoronActiveConcentration")
; Define the source/drain and gate dopant species.
      (define DopSD "ArsenicActiveConcentration")
; Set the background substrate doping concentration.
      (define SubDop 1e18 )
    )
    (begin
; Define the substrate dopant species.
      (define DopSub "ArsenicActiveConcentration")
; Define the source/drain and gate dopant species.
      (define DopSD "BoronActiveConcentration")
; Set the background substrate doping concentration.
      (define SubDop 1e18)
    )
)

; Gate Length parameter imported from Sentaurus Workbench.
(define Lg @lgate@)
; Spacer length used to separate the gate edge from the heavily doped drain implant.
(define Lsp 0.1) ;
; Re-oxidation offset used to place the spacer relative to the gate edge.
(define Lreox 0.01) ;
; Total simulated lateral length, including gate and source/drain extension region.
(define Ltot (+ Lg (* 2 0.01) 0.8) )

; Substrate thickness used for the 2D device cross-section.
(define Hsub 1.0)
; Gate oxide thickness parameter imported from Workbench.
(define Tox @Toxide@)
; Polysilicon gate height.
(define Hpol 0.2)

; Radius used to round the spacer corner for a more realistic geometry.
(define fillet-radius 0.08) ;

; Minimum mesh size control used near critical regions.
(define Gpn 0.005)

; Approximate junction depth used for source/drain implants.
(define Xj 0.1 )

; Derived dimensions are computed from the primary process/device parameters.
;Derived quantities
; Half-device lateral boundary for building a symmetric structure.
(define Xmax (/ Ltot 2.0))
```

```

; Half gate Length.
(define Xg (/ Lg 2.0))
; Spacer edge location measured from the channel center.
(define Xsp (+ Xg Lsp)) ;

; Bottom substrate coordinate.
(define Ysub Hsub)
; Gate oxide top/bottom coordinate based on oxide thickness.
(define Ygox (* Tox -1.0))
; Top polysilicon coordinate.
(define Ypol (- Ygox Hpol))

(define Lcont (- Xmax Xsp)) ;

(define Lcont Xmax)

; Set the default Boolean operation for overlapping SDE geometry creation.
(sdegeo:set-default-boolean "ABA")

; Create a 2D rectangular material region such as silicon, oxide, spacer, or polysilicon.
(sdegeo:create-rectangle
  (position 0.0 Ysub 0.0 )
  (position Xmax 0.0 0.0 )
; Name the silicon body/substrate region.
  "Silicon" "R.Substrate"
)

(sdegeo:create-rectangle
  (position 0.0 0.0 0.0 )
  (position Xsp Ygox 0.0 ) ;
; Name the gate oxide region.
  "Oxide" "R.Gateox"
)

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xsp Ypol 0.0 ) ;
; Define the re-oxidized oxide region around the polysilicon gate.
  "Oxide" "R.PolyReox"
)

(sdegeo:create-rectangle
  (position (+ Xg Lreox) Ygox 0.0 )
  (position Xsp Ypol 0.0 )
; Define the nitride spacer region for LDD/Halo process emulation.
  "Nitride" "R.Spacer"
) ;

(sdegeo:create-rectangle
  (position 0.0 Ygox 0.0 )
  (position Xg Ypol 0.0 )
; Define the polysilicon gate region.
  "PolySi" "R.Polygate"
)

; -rounding spacer

; Round the spacer corner to avoid an unrealistically sharp geometry.
(sdegeo:fillet-2d
  (find-vertex-id (position Xsp Ypol 0.0))
  fillet-radius) ;

; Assign electrical contacts to selected geometry edges.
(sdegeo:set-contact

```

```

    (find-edge-id (position (* (+ Xmax Xsp) 0.5) 0.0 0.0))
    "drain") ;

(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ypol 0.0))
; Label this contact as the gate.
  "gate")
(sdegeo:set-contact
  (find-edge-id (position 5e-4 Ysub 0.0))
; Label this contact as the substrate/body.
  "substrate")

; Define uniform doping profiles for substrate and gate regions.
(sdedr:define-constant-profile "Const.Substrate"
DopSub SubDop )
(sdedr:define-constant-profile-region "PlaceCD.Substrate" "Const.Substrate" "R.Substrate" )

(sdedr:define-constant-profile "Const.Gate"
DopSD 1e20 )
(sdedr:define-constant-profile-region "PlaceCD.Gate" "Const.Gate" "R.Polygate" )

;-----

; Define a geometric window or baseline used for implant placement or mesh refinement.
(sdedr:define-refeval-window "BaseLine.Drain" "Line"
  (position Xsp 0.0 0.0 )
  (position (* Xmax 2.0) 0.0 0.0 ) )

; Define Gaussian implant profiles for source/drain, LDD extension, or halo regions.
(sdedr:define-gaussian-profile "Impl.SDprof"
DopSD
; Set a very high implant peak concentration.
"PeakPos" 0 "PeakVal" 1e20
"ValueAtDepth" SubDop "Depth" Xj
"Gauss" "Factor" 0.4
)

; Source/drain implant profile: high peak concentration for heavily doped drain/source.
(sdedr:define-analytical-profile-placement "Impl.Drain" "Impl.SDprof" "BaseLine.Drain" "Symm" "NoReplace" "Eval")

;-----
;Source/Drain extentions base line definitions
(sdedr:define-refeval-window "BaseLine.DrainExt" "Line"
  (position Xg 0.0 0.0 )
  (position (* Xmax 2.0) 0.0 0.0 ) )

;Source/Drain implant definition
(sdedr:define-gaussian-profile "Impl.SDextprof"
DopSD
; Set a lighter LDD extension peak concentration.
"PeakPos" 0 "PeakVal" 5e18
"ValueAtDepth" SubDop "Depth" (/ Xj 3.0)
"Gauss" "Factor" 0.25
)

;Source/Drain implants
; LDD extension implant profile: Lower concentration and shallower junction than the main drain implant.
(sdedr:define-analytical-profile-placement "Impl.DrainExt" "Impl.SDextprof" "BaseLine.DrainExt" "Symm" "NoReplace" "Eval")

;-----
;Source/Drain base line definitions HALO
(sdedr:define-refeval-window "BaseLine.Halo" "Line"
(position Xg 0.025 0.0)
(position (/ Xj 3.0) 0.025 0.0) )

```

```

;Source/Drain Implant Definition HALO
(sdedr:define-gaussian-profile "Impl.Haloprof"
DopSub
; Set a very high implant peak concentration.
"PeakPos" 0 "PeakVal" 1e20
"ValueAtDepth" 1e18 "Depth" 0.05
"Gauss" "Factor" 0
)

;Source/Drain implants HALO
; Halo implant profile: Local body doping near the junction to suppress punch-through and DIBL.
(sdedr:define-analytical-profile-placement "Impl.Halo" "Impl.Haloprof" "BaseLine.Halo" "Positive" "NoReplace" "Eval")

;-----

; Define mesh refinement sizes for important regions.
(sdedr:define-refinement-size "Ref.Substrate"
(/ Ltot 4.0) (/ Hsub 8.0)
Gpn Gpn )
; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-function "Ref.Substrate" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-region "RefPlace.Substrate" "Ref.Substrate" "R.Substrate" )

(sdedr:define-refeval-window "RWin.Act" "Rectangle"
(position 0.0 0.0 0.0)
(position Xmax (* Xj 1.2) 0.0 ) )
(sdedr:define-refinement-size "Ref.SiAct"
(/ Lcont 4.0) (/ Xj 8.0)
Gpn Gpn )
;-----

; Refine mesh where doping changes rapidly.
(sdedr:define-refinement-placement "Ref.SiAct" "DopingConcentration" "MaxTransDiff" 1)
(sdedr:define-refinement-placement "RefPlace.SiAct" "Ref.SiAct" "RWin.Act" )

(sdedr:define-refinement-size "Ref.GOX"
(/ Ltot 4.0) (/ Tox 4.0)
Gpn (/ Tox 8.0) )
(sdedr:define-refinement-region "RefPlace.GOX" "Ref.GOX" "R.Gateox" )

(sdedr:define-refeval-window "MBWindow.Gate" "Rectangle"
(position (* Xg -1.0) Ypol 0.0)
(position (* Xg 1.0) Ygox 0.0))

; Use multibox refinement to create a fine mesh near the gate/channel and a coarser mesh away from it.
(sdedr:define-multibox-size "MBSIZE.Gate"
(/ Lg 8.0) (/ Hpol 8.0)
(/ Lg 10.0) 2e-4
1.0 -1.35)
(sdedr:define-multibox-placement "MBPlace.Gate" "MBSIZE.Gate" "MBWindow.Gate")

(sdedr:define-refeval-window "MBWindow.Channel" "Rectangle"
(position 0.0 0.0 0.0)
(position (* Xg 1.2) (* 0.5 Xj) 0.0) )
(sdedr:define-multibox-size "MBSIZE.Channel"
(/ Lg 8.0) (/ Xj 8.0)
(/ Lg 10.0) 1e-4
1.0 1.35)
(sdedr:define-multibox-placement "MBPlace.Channel" "MBSIZE.Channel" "MBWindow.Channel")

(sdedr:define-refinement-size "Ref.J"

```

```

(/ 0.016 6.0) 99
(/ 0.016 7.0) 66)

;Gate-Drain Junction

(sdedr:define-refeval-window "RWin.GD" "Rectangle"
  (position (- Xg 0.01) 0.0 0.0)
  (position (+ Xg Lreox) (* Xj 0.2) 0.0) )
(sdedr:define-refinement-placement "RefPlace.GJ" "Ref.J" "RWin.GD" )

;Build Mesh
; Generate the half-device mesh file.
(sde:build-mesh "n@node@_half_msh")

;Reflect the device
; Reflect the half-device structure to build the full symmetric MOSFET and rename the mirrored contact as source.
(system:command "tdx -mtt -x -M 0 -S 0 -ren drain=source n@node@_half_msh n@node@_msh")

;-----

```

11. PROGRAM/ERASE SDEVICE CODE

- Runs transient SONOS program/erase operation.
- Applies pulse waveforms to the gate and models charge trapping in the nitride storage layer.

Annotated Code

```
; File block maps Workbench input/output files for mesh, parameters, plots, currents, and logs.
File {
; Input mesh generated from SDE/SProcess.
    Grid=      "@tdr@"
; Parameter file passed from Sentaurus Workbench.
    Parameter= "@parameter@"
; Output file for terminal currents.
    Current=    "@plot@"
; TDR output file for spatial physical quantities.
    Plot=       "@tdrdat@"
; Log/output file for solver messages.
    Output=     "@log@"
}

; Define terminal boundary conditions and contact properties.
Electrode {
; Gate terminal is initialized and assigned polysilicon/work-function behavior.
    {Name="gate"    Voltage= 0 Material= "PolySi"

; Define a time-dependent voltage waveform at this electrode.
    Voltage=(
    !(
        set t1p 0
; Loop over program/erase cycles specified in Workbench.
        for { set i 1 } { $i <= @cycles@ } { incr i } {
            set t2p [expr $t1p + 1e-6]
            set t3p [expr $t1p + (@dtp@*1e-3)]
            set t4p [expr $t2p + (@dtp@*1e-3)]

            set t1e [expr $t3p + 2.5e-3]
            set t2e [expr $t1e + 1e-6]
            set t3e [expr $t1e + (@dte@*1e-3)]
            set t4e [expr $t2e + (@dte@*1e-3)]

; Generate the program voltage pulse sequence.
            puts "      0 at [format %0.6e $t1p], @Vgp@ at [format %0.6e $t3p], 0 at [format %0.6e $t4p],"
; Generate the erase voltage pulse sequence.
            puts "      0 at [format %0.6e $t1e], @Vge@ at [format %0.6e $t2e], @Vge@ at [format %0.6e $t3e], 0 at [format %0.6e $t4e],"
            set t1p [expr $t3e + 5e-3]
        }

    )!
    )}

; Body/substrate terminal is grounded.
    {Name="substrate" Voltage= 0}
; Drain terminal bias is initialized, then later ramped/swept.
    {Name="drain"    Voltage= 0}
; Source terminal is grounded.
    {Name="source"   Voltage= 0}
}

; Enable global physical models used by SDevice.
Physics {
; Enable electron/hole barrier tunneling through the oxide interfaces.
    eBarrierTunneling "Gateoxb_Substrate"
```

```

; Enable electron/hole barrier tunneling through the oxide interfaces.
hBarrierTunneling "Gateoxb_Substrate"
; Enable electron/hole barrier tunneling through the oxide interfaces.
eBarrierTunneling "Gateoxt_Polygate"
; Enable electron/hole barrier tunneling through the oxide interfaces.
hBarrierTunneling "Gateoxt_Polygate"
}

; Apply silicon-specific recombination and mobility models.
Physics(Material="Silicon") {
; Include Shockley-Read-Hall recombination with doping dependence.
Recombination(SRH(DopingDependence))
; Include doping-dependent mobility and high-field velocity saturation.
Mobility(DopingDependence HighFieldSaturation)
}

; Define trap physics inside the SONOS nitride charge-storage layer.
Physics(Material="NitrideAsSemiconductor") {
; Start definition of donor/acceptor trap levels in the nitride.
Traps(
; Donor-like trap state used to model charge storage.
(Donor Level EnergyMid= 2.5 FromConductionBand
Conc= 1e19
eXSection= 1e-13 hXSection= 1e-13
; Enable electron/hole barrier tunneling through the oxide interfaces.
eBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Enable electron/hole barrier tunneling through the oxide interfaces.
hBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Include field-enhanced emission from traps.
PooleFrenkel
TrapVolume= 1e-15
)
; Acceptor-like trap state used to model charge storage.
(Acceptor Level EnergyMid= 1.0 FromConductionBand
Conc= 1e19
eXSection= 1e-13 hXSection= 1e-13
; Enable electron/hole barrier tunneling through the oxide interfaces.
eBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Enable electron/hole barrier tunneling through the oxide interfaces.
hBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Include field-enhanced emission from traps.
PooleFrenkel
TrapVolume= 1e-15
)
)
}

; Define averaged quantities to record during transient/program/erase simulations.
CurrentPlot {
SpaceCharge(Average(Material="NitrideAsSemiconductor"))
eDensity(Average(Material="NitrideAsSemiconductor"))
hDensity(Average(Material="NitrideAsSemiconductor"))
eTrappedCharge(Average(Material="NitrideAsSemiconductor"))
hTrappedCharge(Average(Material="NitrideAsSemiconductor"))
}

; Select spatial quantities to save for visualization in Sentaurus Visual.
Plot {
eCurrent/Vector hCurrent/Vector
ElectricField
eQuasiFermi hQuasiFermi EffectiveBandgap
ConductionBandEnergy ValenceBandEnergy
Doping SpaceCharge DonorConcentration AcceptorConcentration
SRH Auger
eMobility hMobility
eVelocity hVelocity

```

```

    eParallel hParallel
    eDirectTunnel hDirectTunnel
; Enable electron/hole barrier tunneling through the oxide interfaces.
    eBarrierTunneling hBarrierTunneling
    eTrappedCharge hTrappedCharge
    NonLocal
}

; Configure numerical methods and nonlocal tunneling paths.
Math {
; Define a nonLocal tunneling path across an oxide interface.
    NonLocal "Gateoxb_Substrate" (
; Specify the material interface used for nonLocal tunneling calculation.
        RegionInterface="R.Gateoxb/R.Substrate"
        Length= 9e-7
        Digits= 4
        EnergyResolution= 1e-4
    )
; Define a nonLocal tunneling path across an oxide interface.
    NonLocal "Gateoxt_Polygate" (
; Specify the material interface used for nonLocal tunneling calculation.
        RegionInterface="R.Gateoxt/R.Polygate"
        Length= 5e-7
        Digits= 4
        EnergyResolution= 1e-4
    )
*    CNormPrint
    Iterations= 25
; Use Backward Euler transient integration for numerical stability.
    Transient= BE
}

; Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
# initial solution
; Initialize nitride traps as empty before program/erase transient simulation.
    Set(TrapFilling= Empty)
; Solve Poisson and transport equations self-consistently.
    Coupled (Iterations= 100) { Poisson }
; Solve Poisson and transport equations self-consistently.
    Coupled (Iterations= 100) { Poisson Electron}
; Solve Poisson and transport equations self-consistently.
    Coupled (Iterations= 100 Digits= 8) { Poisson Electron Hole }

; Save spatial solution snapshots for visualization.
    Plot(FilePrefix="n@node@_Zero")

; Allow traps to dynamically fill/empty during the transient simulation.
    UnSet(TrapFilling)

; Perform time-dependent simulation for program/erase pulse or Id-Vg measurement.
    Transient(
        InitialTime= 0 FinalTime= !(puts [expr $t1p])!
; Control numerical step size for convergence and resolution.
        InitialStep= 1e-5 Minstep= 1.e-10 MaxStep= 5e-3
; Solve Poisson and transport equations self-consistently.
    ){ Coupled { Poisson Electron Hole }

    !(
; Save spatial solution snapshots for visualization.
        puts "Plot(FilePrefix=\"n@node@_programmed\" Time=([format %0.6e [expr $t3p]]))"
; Save spatial solution snapshots for visualization.
        puts "Plot(FilePrefix=\"n@node@_programmed_no_bias\" Time=([format %0.6e [expr $t1e]]))"
; Save the programmed/erased state for later electrical characterization.
        puts "Save(FilePrefix=\"n@node@_programmed\" Time=([format %0.6e [expr $t1e]]))"
    )
}

```

```
; Save spatial solution snapshots for visualization.
  puts "Plot(FilePrefix=\"n@node@_erased\" Time=([format %0.6e [expr $t3e]]))"
; Save spatial solution snapshots for visualization.
  puts "Plot(FilePrefix=\"n@node@_erased_no_bias\" Time=([format %0.6e [expr $t1p]]))"
; Save the programmed/erased state for later electrical characterization.
  puts "Save(FilePrefix=\"n@node@_erased\" Time=([format %0.6e [expr $t1p]]))"
    )!
  }
}
```

12. SPROCESS IV sdevice

- Runs SONOS Id-Vg characterization after loading a saved memory state.
- Extracts electrical behavior after programmed or erased charge states.

Annotated Code

```
; File block maps Workbench input/output files for mesh, parameters, plots, currents, and logs.
File {
; Input mesh generated from SDE/SProcess.
    Grid=      "@tdr@"
; Parameter file passed from Sentaurus Workbench.
    Parameter= "@parameter@"
; Output file for terminal currents.
    Current=    "@plot@"
; TDR output file for spatial physical quantities.
    Plot=       "@tdrdat@"
; Log/output file for solver messages.
    Output=     "@log@"
}

; Define terminal boundary conditions and contact properties.
Electrode {
; Gate terminal is initialized and assigned polysilicon/work-function behavior.
    {Name="gate"      Voltage= 0   Material = "PolySi"
; Define a time-dependent voltage waveform at this electrode.
    Voltage=(0 at 0, -1.25 at 1e-6, -1.25 at 2e-6, 3.25 at 1)
    }
; Body/substrate terminal is grounded.
    {Name="substrate" Voltage= 0}
; Drain terminal bias is initialized, then later ramped/swept.
    {Name="drain"      Voltage= 0   Voltage=(0 at 0, 1 at 1e-6)}
; Source terminal is grounded.
    {Name="source"     Voltage= 0}
}

; Enable global physical models used by SDevice.
Physics {
; Enable electron/hole barrier tunneling through the oxide interfaces.
    eBarrierTunneling "Gateoxb_Substrate"
; Enable electron/hole barrier tunneling through the oxide interfaces.
    hBarrierTunneling "Gateoxb_Substrate"
; Enable electron/hole barrier tunneling through the oxide interfaces.
    eBarrierTunneling "Gateoxt_Polygate"
; Enable electron/hole barrier tunneling through the oxide interfaces.
    hBarrierTunneling "Gateoxt_Polygate"
}

; Apply silicon-specific recombination and mobility models.
Physics(Material="Silicon") {
; Include Shockley-Read-Hall recombination with doping dependence.
    Recombination(SRH(DopingDependence))
; Include doping-dependent mobility and high-field velocity saturation.
    Mobility(DopingDependence HighFieldSaturation)
}

; Define trap physics inside the SONOS nitride charge-storage layer.
Physics(Material="NitrideAsSemiconductor") {
; Start definition of donor/acceptor trap levels in the nitride.
    Traps(
; Donor-like trap state used to model charge storage.
        (Donor Level EnergyMid= 2.5 FromConductionBand
        Conc= 1e19
```

```

        eXSection= 1e-13 hXSection= 1e-13
; Enable electron/hole barrier tunneling through the oxide interfaces.
        eBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Enable electron/hole barrier tunneling through the oxide interfaces.
        hBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Include field-enhanced emission from traps.
        PooleFrenkel
        TrapVolume= 1e-15
    )
; Acceptor-like trap state used to model charge storage.
    (Acceptor Level EnergyMid= 1.0 FromConductionBand
    Conc= 1e19
    eXSection= 1e-13 hXSection= 1e-13
; Enable electron/hole barrier tunneling through the oxide interfaces.
        eBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Enable electron/hole barrier tunneling through the oxide interfaces.
        hBarrierTunneling(NonLocal= "Gateoxb_Substrate" NonLocal= "Gateoxt_Polygate")
; Include field-enhanced emission from traps.
        PooleFrenkel
        TrapVolume= 1e-15
    )
}

; Define averaged quantities to record during transient/program/erase simulations.
CurrentPlot {
    SpaceCharge(Average(Material="NitrideAsSemiconductor"))
    eDensity(Average(Material="NitrideAsSemiconductor"))
    hDensity(Average(Material="NitrideAsSemiconductor"))
    eTrappedCharge(Average(Material="NitrideAsSemiconductor"))
    hTrappedCharge(Average(Material="NitrideAsSemiconductor"))
}

; Select spatial quantities to save for visualization in Sentaurus Visual.
Plot {
    eCurrent hCurrent
    ElectricField
    eQuasiFermi hQuasiFermi EffectiveBandgap
    ConductionBandEnergy ValenceBandEnergy
    Doping SpaceCharge DonorConcentration AcceptorConcentration
    SRH Auger
    eMobility hMobility
    eVelocity hVelocity
    eParallel hParallel
    eDirectTunnel hDirectTunnel
; Enable electron/hole barrier tunneling through the oxide interfaces.
    eBarrierTunneling hBarrierTunneling
    eTrappedCharge hTrappedCharge
    NonLocal
}

; Configure numerical methods and nonlocal tunneling paths.
Math {
; Define a nonlocal tunneling path across an oxide interface.
    NonLocal "Gateoxb_Substrate" (
; Specify the material interface used for nonlocal tunneling calculation.
        RegionInterface="R.Gateoxb/R.Substrate"
        Length= 9e-7
        Digits= 4
        EnergyResolution= 1e-4
    )
; Define a nonlocal tunneling path across an oxide interface.
    NonLocal "Gateoxt_Polygate" (
; Specify the material interface used for nonlocal tunneling calculation.
        RegionInterface="R.Gateoxt/R.Polygate"
        Length= 5e-7
    )
}

```

```

    Digits= 4
    EnergyResolution= 1e-4
)
* CNormPrint
  Iterations= 25
; Use Backward Euler transient integration for numerical stability.
  Transient= BE
}

; Solver sequence: first build a stable equilibrium solution, then apply bias ramps/sweeps.
Solve {
  Transient (
    InitialTime= 0 FinalTime= 1e-6
    ; Control numerical step size for convergence and resolution.
    InitialStep= 1e-8 MaxStep= 5e-7 Minstep= 1.e-10
    ; Solve Poisson and transport equations self-consistently.
  ){ Coupled {Poisson Electron Hole}
; Load a previously saved SONOS programmed or erased state.
  Load(FilePrefix="n@previous|-1@_@SONOSstate@" Time=(0))
}

; Set current-output file prefix for this sweep.
NewCurrentPrefix="n@node@_IdVg_"
  Transient (
    InitialTime= 1e-6 FinalTime= 1
    ; Control numerical step size for convergence and resolution.
    InitialStep= 1e-6 MaxStep= 3.125e-02 Minstep= 1.e-7
    ; Solve Poisson and transport equations self-consistently.
  ){ Coupled {Poisson Electron Hole}
; Sample terminal currents during the sweep.
    CurrentPlot( Time=(Range=(0 1) Intervals= 32) )
  }
}

```